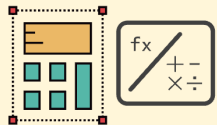


Input Layer

Task Battery



Intuitive Tasks
(e.g., MMLU)



Analytical Tasks
(e.g., GSM8K)



Conflict Tasks
(e.g., TruthfulQA)

Task Battery
(N=18,565 Items)

Processing Layer

Dual-Process Simulation



System 1: Fast & Intuitive



Model: GPT-4o-mini



High
Temp
(0.9)

Prompt:
Zero-shot/
Gut Feeling

instant response



System 2: Slow & Analytical

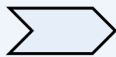


Model: GPT-4o

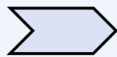


Low
Temp
(0.2)

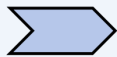
Prompt:
Chain-of-
Thought(CoT)



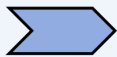
(1)
Analyze



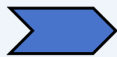
(2)
Identify
Traps



(3)
Apply
Principles



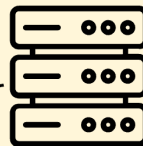
(4)
Reason
Steps



(5)
Verify

Execution Layer

Procedural Framework



Parallel API
Framework
Concurrency Limits
(S1:20 / S2:10)



3 Independent
Rounds
(Randomized)

Output Layer

Data Collection & Metrics



Accuracy



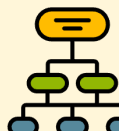
Confidence Score



Response Time



Token Usage



Reasoning Steps